

Charge transport and diffusion of ionic liquids in nanoporous silica membranes

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Charge transport in 1-hexyl-3-methylimidazolium hexafluorophosphate ionic liquid in oxidized nanoporous silicon membranes is investigated in a wide frequency and temperature range by a combination of Broadband Dielectric Spectroscopy (BDS) and Pulsed Field Gradient Nuclear Magnetic Resonance (PFG NMR). By applying the Einstein–Smoluchowski relations to the dielectric spectra, diffusion coefficient is obtained in quantitative agreement with independent PFG NMR measurements. More than 10-fold systematic decrease in the effective diffusion coefficient from the bulk value is observed in hydrophilic silica nanopores. A model assuming a reduced mobility at the pore–matrix interface is shown to provide a quantitative explanation for the remarkable decrease of effective transport quantities (such as diffusion coefficient, dc conductivity and consequently, the dielectric loss) of the ionic liquid in non-silanized membranes. This approach is supported by the observation that silanization of porous silica membranes results in a significant increase of the effective diffusion coefficient, which approaches the value for the bulk liquid.

1. Introduction

Nanopores have a great potential for use in drug delivery, nano-fluidic transistors, DNA translocation, catalysis, sensors for chemical agents, nanobatteries, supercapacitors, templates for nanoparticle self-assembly as well as in fundamental research.^{1–5} These applications require in-depth understanding of the motion of molecules or charge carriers inside the pores. One research direction has been geared towards employing nanopores in power sources and storage devices.⁶ The huge capacitances that are realized in such materials make them particularly suitable for replacing the current generation of supercapacitors. Due to their interesting properties such as high conductivity, electrochemical stability, negligible vapour pressure, thermal stability and wide liquid ranges, ionic liquids (ILs) are under investigation as more efficient electrolytes for nano-capacitors and fuel cells.^{7–11} The combination of the properties of the nanopores and ionic liquids could be ideal for the optimization of these devices. Ionic liquids are being investigated for use as electrolytes in power sources such as fuel cells and batteries. This has been triggered by the high dc conductivity (among other factors) observed in these materials. The decrease in the transport parameters (such as diffusion coefficient and dc conductivity) obtained in SiO₂ membranes is not desirable.

Previous studies have focused on the synthesis and characterization of porous glasses ranging from those with complex geometries such as disordered Vycor and M41S silica materials^{12,13} to the well-ordered hexagonally packed alumina nanopores.^{14–16} On the other hand, there is a concerted effort to understand the mechanism of charge transport in ionic liquids,^{11,17–19} but a number of questions remain unanswered. For instance, it is

necessary to investigate the effect of the interface between the sample and the pore wall on transport quantities such as conductivity, diffusivity, viscosity and dielectric loss, among others.

In this article, we combine broadband dielectric spectroscopy (BDS) and pulsed field gradient nuclear magnetic resonance (PFG NMR) to probe charge transport in 1-hexyl-3-methylimidazolium hexafluorophosphate ionic liquid in porous silica membranes consisting of pores of average diameter 7.5 nm. The combination of both techniques allows the measurement in a much broader temperature range than PFG NMR alone, covering 10 decades of diffusivity. Over 10-fold decrease in the diffusion coefficient of IL in the hydrophilic pores is found as compared to the bulk value. Silanization reverses this trend, showing that the interaction of the ionic liquid with the pore surface plays a fundamental role in the observed confinement effects.

2. Experimental details

The IL investigated in this study (1-hexyl-3-methylimidazolium hexafluorophosphate, [HMIM][PF₆], molecular weight = 312.24 g mol^{−1}, purity > 99%) was purchased from Iolitec GmbH and used as received. Membranes of porous silicon with one-dimensional channels—prepared by electrochemical etching of highly doped p-type ⟨100⟩ oriented silicon substrate with resistivity less than 0.005 Ω cm—were used. The electrolyte contained HF (48%) and C₂H₅OH in ratio (volume) of 1 : 1. Anodization current density $j = 10 \text{ mA cm}^{-2}$ was applied. The porous membranes were obtained after a final electro-polishing step with $j = 700 \text{ mA cm}^{-2}$ applied for 3–4 s. The resulting porous silicon (PS) membranes (about 50 ± 2 μm thick) were thereafter thermally oxidized at 1100 K for 3 h.^{20–22} The insulating oxidized membranes (glassy SiO₂ membranes) were annealed in an evacuated

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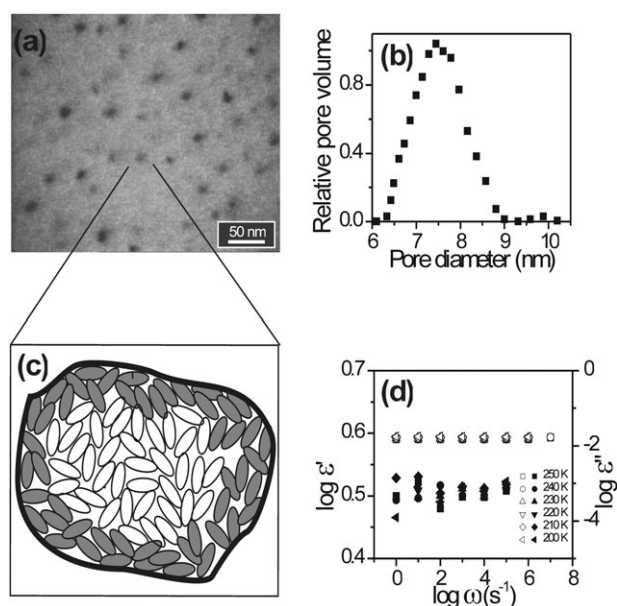


Fig. 1 (a) Scanning Electron Microscopy analysis (SEM): bottom view of SiO₂ membrane, (b) pore size distribution determined by NMR-cryoporometry,²³ (c) schematic representation of one nanopore filled with ionic liquid (the dark region represents the adsorption layer while the remaining region is pictured as exhibiting bulk-like properties), (d) the dielectric response of the empty SiO₂ membrane: ϵ' is the real part (open symbols) and ϵ'' the imaginary part (solid symbols) of the complex dielectric function *versus* frequency at different temperatures.

chamber (10^{-5} mbar) at 470 K for 24 h prior to filling with the ionic liquid by capillary wetting (for a period of 48 h at 300 K). For characterization of the nanopores, scanning electron microscopy (SEM) and NMR-cryoporometry measurements²³ were performed. The results are as shown in Fig. 1(a) and (b). Using both techniques, the average pore diameter for the oxidized membranes is found to be about 7.5 ± 0.7 nm. To determine the porosity, Barrett–Joyner–Halenda (BJH) analysis method²⁴ was applied to the nitrogen sorption results obtained at 77 K by means of BelSorp-Mini II (BEL Japan Inc.). For SiO₂ membranes, porosity of 7% was found.

In order to study the role of surface interaction on charge transport in ionic liquids, two types of SiO₂ membranes were prepared: untreated and treated (silanized) SiO₂ membranes. It is expected that an adsorption layer can be formed between the silica surface and the IL in the nanopore (illustrated in Fig. 1(c)) for the non-silanized case.²⁵ The untreated surface of silica membrane is densely occupied by non-hydrogen bonded silanol groups. In order to reduce the influence of this layer on the measured transport properties, the surface was altered by silanization. For this modification, hexamethyldisilazane (HMDS) (purity $\geq 99.0\%$, purchased from Sigma-Aldrich) was injected into the vacuum preparation chamber containing the SiO₂ membrane at 350 K. After 30 min, the chamber with silica membrane was again evacuated for 3 h at 350 K. By treating the membrane with HMDS, the silanol groups were replaced with trimethylsilyl groups. The inner surface consequently became hydrophobic. The infrared spectrum of the modified surface indicated that the exchange of silanol groups was complete. Fig. 2 shows the absorption coefficient in the IR

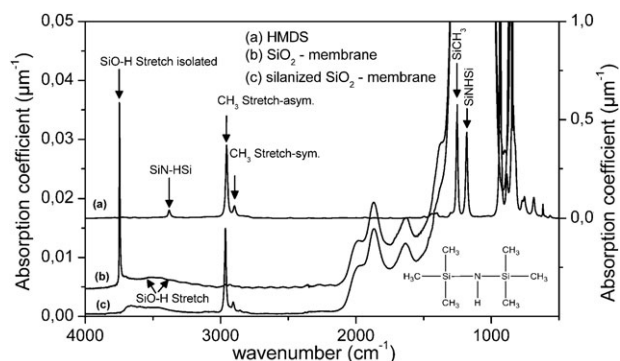


Fig. 2 Infrared absorption coefficients of (a) hexamethyldisilazane (HMDS) (right y -axis), (b) untreated SiO₂ membrane consisting of pores of average diameter 7.5 nm and (c) modified membrane by silanization with HMDS (left y -axis). In the spectrum of the modified SiO₂ membrane, the vibration of the free O–H is not present but the CH₃ stretching vibrations are observed instead. Contrary to pure HMDS, there are no vibrations of N–H in the modified silica membrane. The structure of HMDS is included in the inset.

range of HMDS, silanized and non-silanized SiO₂ membranes. The untreated silica membrane exhibits a sharp band at ~ 3740 cm^{−1} corresponding to non-hydrogen-bonded OH groups. The intensity of this band dramatically decreases after the membrane is treated with HMDS proving that the exchange of silanol groups is complete. The C–H stretching bands in the region 2800–3000 cm^{−1} verify the presence of methyl groups.

The concentration of CH₃ groups in the silanized membrane can be calculated from the IR absorption spectra using the Lambert–Beer law.²⁶ As a reference system we chose HMDS, where the concentration of methyl groups is known and equal to 6[HMDS]. Therefore

$$[\text{CH}_3] = 6 \frac{\int \alpha_{\text{membrane}}}{\int \epsilon_{\text{HMDS}}} = 6[\text{HMDS}] \frac{\int \alpha_{\text{membrane}}}{\int \alpha_{\text{HMDS}}} \quad (1)$$

where: α is the absorption coefficient and ϵ the molar absorption coefficient. The integrals show that the whole area under the C–H band is considered. Substitution of the experimental values gives $[\text{CH}_3] = 6 \times \frac{0.76 \text{ g cm}^{-3}}{161.4 \text{ g mol}^{-1}} \times 0.033 = 0.93 \text{ M}$. The distance between silane groups can be estimated by $[\text{CH}_3] \sim \frac{3}{N_A} \frac{4p}{d} \frac{1}{d_{\text{silane}}^2}$, where p is the porosity and d is the pore diameter, since the pore surface is much larger than the membrane outer surface. Using $p = 0.07$ and $d = 7.5$ nm, we get $d_{\text{silane}} = 0.45$ nm. For comparison, the Si–Si distance in the HMDS molecule is 0.304 nm whereas the distance between methyl groups bonded to different Si atoms ranges from 0.38 to 0.53 nm. Therefore, we prove that the pore surface is fully silanized and covered by a hydrophobic layer with thickness ≈ 0.5 nm (geometry determined by the ArgusLab software).

In the current article, we investigate charge transport of [HMIM][PF₆] in bulk and in nanopores by two independent techniques, namely pulsed field gradient nuclear magnetic resonance (PFG NMR) and broadband dielectric spectroscopy (BDS). PFG NMR diffusion measurements are based on alternating radio and magnetic field gradient pulses applied to the sample nuclei.^{27–29} Applying magnetic field gradient

pulses, the spin echo becomes sensitive to the translational motion of the molecules, which carry the nuclear spin under study. Assuming that the PFG is applied along the z axis on the laboratory frame of reference (parallel to the pores), the diffusive motion of the molecules leads to an attenuation of the spin echo intensity that is dependent on the field gradient strength g , pulse duration δ , observation time t , and the self-diffusion coefficient D_z . Thus, in the case of the normal unrestricted self-diffusion in a bulk fluid, the spin echo intensity decays exponentially with respect to the square of the magnetic field gradient strength g^2 ,

$$\frac{M(\delta g, t)}{M(\delta g = 0, t)} = \exp\{-(\gamma \delta g)^2 D_z t\} \quad (2)$$

where γ is the gyromagnetic ratio of the observed type of nucleus. D_z can thus be obtained from the time dependence of the spin echo attenuation. The measurements presented in this article were performed on a 400 MHz spectrometer with a home-built gradient unit with gradients up to 35 T m⁻¹. The stimulated echo sequence has been utilized with an observation time of 10 ms. Both proton and fluoride nuclei were probed. The presented diffusion coefficients are the sum of the cation and anion respective values.

Dielectric measurements were carried out using a Novocontrol high-resolution Alpha Dielectric Analyzer (0.1 Hz–10 MHz) in the temperature range 170 K–380 K, aided by Quattro temperature controllers with temperature stability better than 0.2 K, under a dry nitrogen atmosphere. All measurements were conducted using platinum electrodes. For dielectric measurements of ILs in nanoporous membranes, it is imperative that the membranes are insulating. The dielectric spectra show low dielectric loss, $\epsilon''_{\text{SiO}_2} \leq 10^{-3}$ (Fig. 1d), implying that the conductivities are lower than 10⁻¹⁵ S cm⁻¹. The SiO₂ membranes used in the current study are thus practically non-conducting.

3. Results and discussion

Due to its unique ability to probe molecular fluctuations and charge transport over a broad frequency and temperature range, broadband dielectric spectroscopy (BDS) has proven indispensable in the quest to understand the underlying mechanisms of charge transport and dynamic glass transition in ion conducting glass-forming systems.^{11,18,25} BDS measures the complex dielectric function, $\epsilon^*(\omega) = \epsilon'(\omega) - i\epsilon''(\omega)$, which is equivalent to the complex conductivity function, $\sigma^*(\omega) = \sigma'(\omega) + i\sigma''(\omega)$. This is expressed as $\sigma^*(\omega, T) = i\epsilon_0\omega\epsilon^*(\omega, T)$, implying that $\sigma' = \epsilon_0\omega\epsilon''$ and $\sigma'' = \epsilon_0\omega\epsilon'$ (ϵ_0 being the vacuum permittivity, ω the radial frequency, T the temperature). Since the two forms emphasize different facets of the same process, it is instructive to present both as shown in Fig. 3. The dielectric spectra of the ionic liquid [HMIM][PF₆] are given for measurements in bulk and non-silanized SiO₂ nanopores. The real part of the conductivity σ' in the intermediate frequencies exhibits a plateau, whose value yields the dc conductivity, σ_0 . The decrease observed at lower frequencies is known to be due to electrode polarisation.^{19,25} At higher frequencies, dispersion sets in at a frequency corresponding to the characteristic rate ω_c (which is attributed to the cross-over from diffusive

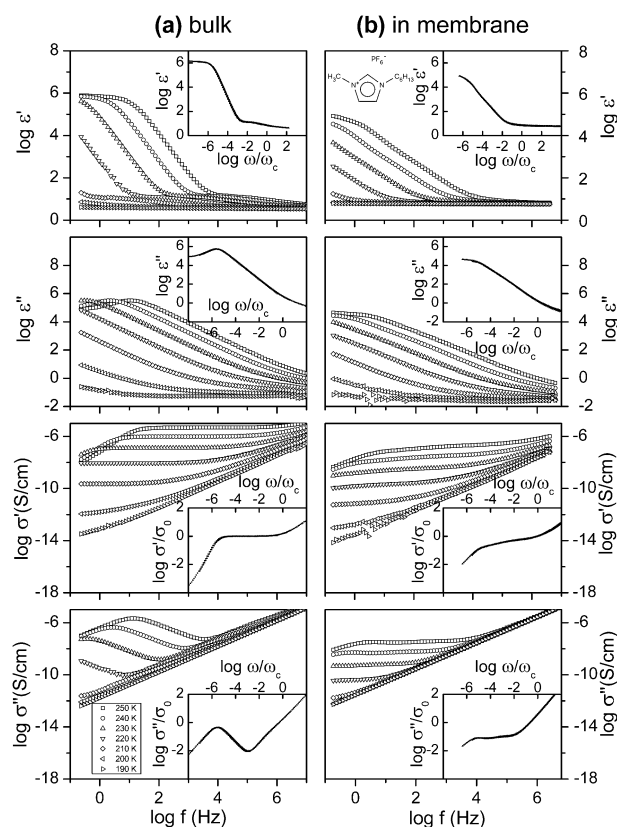


Fig. 3 (a) Complex dielectric function ($\epsilon^*(\omega) = \epsilon'(\omega) - i\epsilon''(\omega)$) and complex conductivity function ($\sigma^*(\omega) = \sigma'(\omega) + i\sigma''(\omega)$) of 1-hexyl-3-methylimidazolium hexafluorophosphate [HMIM][PF₆] versus radial frequency at different temperatures as indicated for the bulk ionic liquid. Insets: scaling with respect to the characteristic rate ω_c and in case of the conductivity spectra, to the dc conductivity, σ_0 . (b) The complex dielectric function and complex conductivity function of [HMIM][PF₆] versus radial frequency at different temperatures in untreated SiO₂ membrane. Insets: scaling with respect to the characteristic rate ω_c and in case of conductivity spectra, to the dc conductivity σ_0 . The structure of [HMIM][PF₆] is included. The error bars are smaller than the size of the symbols, if not specified otherwise.

to sub-diffusive dynamics) and σ' shows a power-law dependence. Superposition of the experimental data by scaling with respect to ω_c and σ_0 at different temperatures yields single curves as shown in the insets of Fig. 3a and b for the measurements in bulk and untreated nanopores respectively. This demonstrates that the underlying mechanisms of charge transport and electrode polarisation follow an *identical* type of thermal activation in the entire temperature range.^{18,30} This implies that the low loss of the silica matrix influences the obtained results. However, the curves are different due to the complicated geometry of electrical contacts at the entrance of the nanopores, which alters the polarization contribution.

As displayed in Fig. 4, the temperature dependence of the characteristic rate $\omega_c(T)$ of the [HMIM][PF₆] in bulk in the untreated (non-silanized) and treated (silanized) SiO₂ nanopores exhibits a Vogel–Fulcher–Tammann (VFT) dependence, typical for glass-forming liquids. The presentation of plots enables an in-depth understanding of the distinctive trend of the [HMIM][PF₆] in the untreated (Fig. 4(a)) and silanized

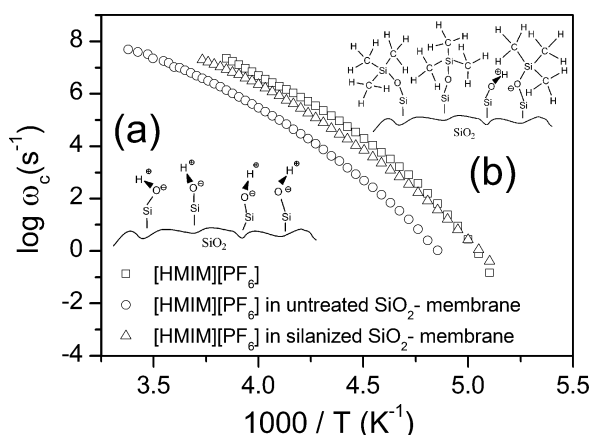


Fig. 4 Temperature dependence of the characteristic (diffusion) rate $\omega_c(T)$ of the [HMIM][PF6] in the bulk, in the non-silanized and silanized SiO_2 nanopores (denoted by squares, circles and triangles respectively). Inset: the chemical structures of (a) untreated and (b) silanized SiO_2 membranes.

SiO_2 membranes (Fig. 4(b)). The characteristic rate in the untreated nanopores is lower by about 1.4 decades than the bulk. After silanization of the SiO_2 membrane with HMDS, it is observed that the characteristic rate ω_c remarkably increases by about one decade compared to untreated SiO_2 membrane, becoming equal—within the experimental uncertainty—to the value for bulk.

Recently,¹⁹ it has been demonstrated that the diffusion coefficient can be determined from BDS spectra by a combination of the Einstein and Einstein–Smoluchowski equations. Within this framework, the dc conductivity is given by

$$\sigma_0 = nq\mu = \frac{nq^2}{kT} D = \frac{nq^2 \lambda^2 \omega_c}{kT} \Rightarrow D = \frac{\lambda^2 \omega_c}{2}, \quad (3)$$

where n denotes the effective number density (given by $N\gamma(T)$ where $\gamma(T)$ is the fraction of mobile charge carriers at a given temperature and N is the total number density of charge carriers available in the material), μ is the mobility of charge carriers, q is the elementary electric charge, D is the diffusion coefficient, k the Boltzmann constant, λ denotes the hopping length and ω_c is the characteristic (diffusion) rate. By assuming hopping lengths of the order of the Pauling diameter *i.e.* $\lambda = 0.18$ nm,¹⁸ diffusion coefficients are readily determined and are also shown in Fig. 5. The self-diffusion coefficients of [HMIM][PF6] have not been studied before, but our results are comparable to those obtained for the similar ionic liquid—1-butyl-3-methylimidazolium hexafluorophosphate [BMIM][PF6]—in bulk. For example, at $T = 293$ K we obtain a value of $1.0 \times 10^{-11} \text{ m}^2 \text{ s}^{-1}$, compared to $9 \times 10^{-12} \text{ m}^2 \text{ s}^{-1}$ (Tokuda *et al.*³¹) and $1.2 \times 10^{-11} \text{ m}^2 \text{ s}^{-1}$ (Umecky *et al.*³²). VFT temperature dependences of the diffusion coefficients are observed both in bulk and nanopores, even though deviations close to the respective glass temperatures are found. Thus, a single activation energy cannot be obtained. The VFT parameters obtained are summarized in Table 1. In comparison to the results of Tokuda *et al.* and Umecky *et al.*, for [BMIM][PF6], the value of the parameter B we found is systematically higher as shown in Table 2. This is because

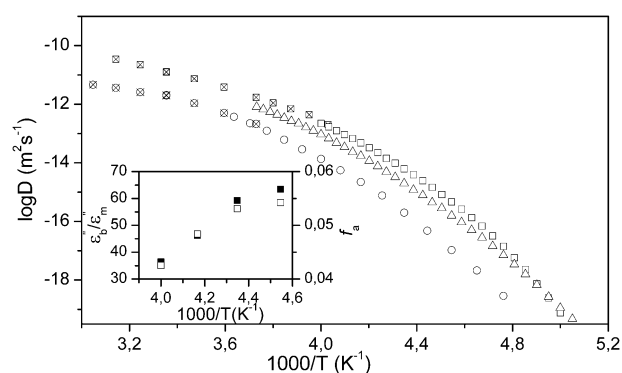


Fig. 5 Diffusion coefficients determined by applying the Einstein–Smoluchowski eqn (3) to the dielectric spectra of [HMIM][PF6] in bulk (open squares), untreated nanopores (open circles) and silanized nanopores (open triangles). The respective values obtained by PFG NMR are shown by crossed symbols. Inset: temperature dependence of the ratio of the dielectric loss in bulk to the total dielectric loss in untreated nanopores (solid symbols). The surface area fractions corresponding to the adsorption layer (f_a) as a function of temperature are also given (open symbols).

Table 1 The temperature dependence of the self-diffusion coefficients approximated by Vogel–Fulcher–Tammann (VFT) equation expressed as $D = D_0 \exp[-B/(T - T_0)]$, where the constants D ($\text{m}^2 \text{ s}^{-1}$), B (K) and T_0 (K) are adjustable parameters. The VFT parameters for the self-diffusion coefficient of [HMIM][PF6] in bulk, non-silanized and silanized nanopores are obtained as indicated

	$\log D_0/\text{m}^2 \text{ s}^{-1}$	B/K	T_0/K
Bulk	-7.4 ± 0.2	2530 ± 50	159 ± 1
Non-silanized pores	-7.9 ± 0.3	3000 ± 200	161 ± 3
Silanized pores	-6.3 ± 0.3	3800 ± 200	146 ± 5

Table 2 Vogel–Fulcher–Tammann parameters for the diffusivity of bulk [BMIM][PF6] (Tokuda *et al.*³¹ and Umecky *et al.*³²) for comparison with the current results (see Table 1)

	$\log D_0/\text{m}^2 \text{ s}^{-1}$	B/K	T_0/K
Tokuda <i>et al.</i>	-5.49 ± 0.5	1032 ± 44	167 ± 3
Umecky <i>et al.</i>	-7.66	879 ± 44	183

our diffusivities span 6 decades more than those in the mentioned references. It is well-established that the degree of deviation from the Arrhenius-type temperature dependence in glass-forming liquids becomes more pronounced at lower temperatures.²⁵ Therefore, the apparent disparity in the value of B obtained is expected.

In order to gain deeper insight into the mechanisms of charge transport in ionic liquids under geometrical confinement, PFG NMR, a well-established technique for probing bulk diffusion, is used for measurements both in bulk and nanopores. The results from the two methods coincide at temperatures much higher than the glass transition. In addition, the applicability of the superposition principle shows that the diffusion mechanism remains the same down to the T_g . Therefore, BDS can extend the temperature range of PFG NMR, allowing coverage of 10 decades of diffusivity. The remarkable quantitative agreement of the diffusion coefficients measured by BDS and PFG NMR measurements under conditions of nano-confinement is reported for the first time.

The diffusion coefficient in the hydrophilic, untreated, membrane decreases by about one decade compared to the bulk IL. Notably, this decrease holds on the time scales from nanoseconds to tens of seconds, probed by both BDS and PFG NMR. On the other hand, the IL in the silanized membrane shows only small deviations from the bulk values. In general, two effects may contribute to the slowing down of the diffusivity of confined molecules, namely the interaction with the pore walls and the so-called tortuosity effect.³³ In what follows, we analyze their possible impact in our experiments.

There is sufficient evidence in the literature that even very carefully grown linear pores in porous silicon possess a certain degree of mesoscale disorder.^{34,35} One of the consequences is the distribution of pore sizes shown in Fig. 1(b). Presumably, this distribution reflects the variation of the pore diameter along the pore axis. In turn, molecular propagation in such a disordered structure can be slowed down^{33,36} and the reduction of molecular diffusivities can be easily assessed by computer simulations. To estimate the disorder effect upon molecular diffusivities, Monte Carlo simulations of a point-like particle in linear pores composed of small, statistically distributed cylindrical sections with different pore diameters are employed. Distribution $P(d)$ of pore diameters is taken to be Gaussian

$$P(d) \propto \exp\left\{-\frac{(d-d_0)^2}{2\Sigma^2}\right\} \quad (4)$$

where d_0 is the mean pore diameter and Σ the standard deviation. In the simulations, the mean square displacements $\langle(z-z_0)^2\rangle$ along the pore axis direction are followed as function of Monte Carlo steps number n . At sufficiently large n , the function $\langle(z-z_0)^2\rangle$ was found to grow proportionally to n , as expected for random processes. Thereafter, using the Einstein relation $\langle(z-z_0)^2\rangle = 2Dn$, the diffusivity D can easily be calculated. The normalized diffusivities D/D_0 , where D_0 is the bulk diffusivity, for different structures with different pore size distributions (*i.e.* different values Σ in eqn (4)) are evaluated. The results obtained show that with the experimentally estimated value of $\Sigma = 0.26$, the diffusivity is expected to decrease only slightly compared to D_0 (by $\sim 5\%$), suggesting that geometrical disorder is of no importance in the explanation of the results in Fig. 5.

The ability of hydrogen-bonded liquids to attach to untreated SiO_2 surface (due to strong hydrophilicity of the walls of SiO_2 membranes) is well established from previous studies of the role of hydrogen-bond (H-bond) formation on molecular dynamics.^{25,36–38} In such cases, surface bound layer(s) have to be expected (Fig. 1(c)) and the remaining volume is filled with bulk-like molecules—an effect not yet investigated for ionic liquids. Based on this consideration, the measured dielectric response can be modelled in terms of three contributions; “bulk-like” molecules in the pores, adsorbed molecules at the pore–matrix interface, and the insulating porous silica matrix. The simple model consists of the three impedances in parallel (Z_b^* —bulk impedance, Z_a^* —impedance of the adsorption layer and $Z_{\text{SiO}_2}^*$ —impedance of silicon dioxide matrix) corresponding to the three main components. The total measured impedance Z_m^* is given by $Z_m^* = \frac{1}{i\omega \epsilon_m^* C_m}$ where ϵ_m^* represents the measured complex dielectric function and C_m

denotes the vacuum capacitance of the measurement cell. Thus, the real and imaginary parts of the measured dielectric function can be expressed as

$$\epsilon_m' = f_b \epsilon_b' + f_a \epsilon_a' + f_{\text{SiO}_2} \epsilon_{\text{SiO}_2}' \quad (5a)$$

and

$$\epsilon_m'' = f_b \epsilon_b'' + f_a \epsilon_a'' + f_{\text{SiO}_2} \epsilon_{\text{SiO}_2}'' \quad (5b)$$

where ϵ_b^* , ϵ_a^* , $\epsilon_{\text{SiO}_2}^*$ represent the complex dielectric functions of bulk, adsorption layer and silica membrane, respectively. f_b , f_a and f_{SiO_2} represent the surface area fractions of the bulk liquid, adsorption layer and SiO_2 matrix. Further insight can be gained by considering eqn (5) in detail. With porosity of 7%, one obtains $f_b + f_a = 0.07$. The dielectric response from the empty untreated SiO_2 membrane ϵ_{SiO_2}' is given in Fig. 1(d). The oxidized SiO_2 membrane is observed to be insulating with $\epsilon_{\text{SiO}_2}'' \leq 10^{-3}$. It is therefore evident that $\epsilon_{\text{SiO}_2}'' < 10^{-3} \Rightarrow f_{\text{SiO}_2} \epsilon_{\text{SiO}_2}'' < 0.01$ and $\frac{\epsilon_b''}{\epsilon_m''} \cong \frac{1}{f_b}$ (where the adsorbed layer is expected to be much less conducting, *i.e.* $\epsilon_a'' \ll \epsilon_b''$). From the analysis of eqn (5) based on these simple considerations, the following experimental features are obtained: (i) the total measured dielectric loss (and consequently, characteristic charge transport rate and diffusion coefficient) in nanopores is about 1.4 decades less than the bulk value, (ii) ϵ_m' is dominated by the contribution from the SiO_2 membrane, thereby leading to the absence of the step at frequency corresponding to ω_c as compared to the bulk spectra, (iii) the deviation of transport quantities (diffusion coefficient, charge transport rate ω_c , dc conductivity, dielectric loss) from the bulk value increases with decreasing temperature. Thus, $f_a(T_1) < f_a(T_2)$ if $T_1 > T_2$, in quantitative agreement with the values of the size of the adsorption layer at different temperatures as shown in the inset of Fig. 5. With decreasing temperature, the bulk-like contribution from the pores dramatically decreases.²⁸ The lack of H-bonding and the decreasing dipole–dipole interactions explain the increased diffusivity in the silanized nanopores. Recent studies³⁹ on a series of alkanes–fatty alcohols illustrate the role of dipole–dipole interactions on diffusivity. The conductivity in the treated pores is much higher, in accordance with the fact that the diffusion coefficient increases. Calculating the dielectric loss ratio $\epsilon_{\text{bulk}}''/\epsilon_{\text{treated}}''$ gives a value of ~ 8 in the entire temperature range. This would lead to a negative value for f_a , implying that there is no adsorbed layer in the treated membranes, which also means that the conductivity is higher than in the bulk.

Understanding the mechanism of the formation and the extent of the adsorption layer as well as its temperature dependence enables one to devise means of eliminating (or countering) it. Thus, we proceed to modify the surface of the SiO_2 membrane by silanization in order to probe the nature of the adsorbed layer. It turns out that silanization of SiO_2 pore walls significantly reduces this effect (see Fig. 5). This finding is easily understood by considering our model (eqn (5)). Due to the absence of hydrogen bonding between the ionic liquid and the membrane, we expect that the adsorption layer is both thinner and more mobile, compared to the untreated membrane case.

4. Conclusions

Charge transport in ionic liquids confined in nanoporous silica membranes is investigated by BDS and PFG NMR. Quantitative agreement between the diffusion coefficients both in bulk and nanopores as determined by PFG NMR and BDS is obtained. The differences of diffusivity in the pores compared to the bulk are explained within a theoretical framework that considers, in addition to the bulk response, the contribution from the layer of reduced mobility at the pore–matrix interface. A 10-fold decrease in the untreated, hydrophilic membrane is observed, due to hydrogen bonding between the ionic liquid and the silanol groups. In agreement with the model presented, a remarkable increase of the diffusion coefficient of the ionic liquid in nanopores is observed upon silanization of the silica membranes.

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